

COMPUTER NETWORKS · UNIT 1

Introduction to Computer Networks

SHRABANTI DE (COLLEGEDEKHO)



01 TABLE OF CONTENTS

What You'll Learn in Unit 1

Six chapters covering everything from network basics to architecture.

CHAPTER 01

01

**Network Foundations**

Definition, history, uses & applications of computer networks.

CHAPTER 02

02

**Network Hardware & Servers**

Workstations, servers and their types in a network network.

CHAPTER 03

03

**Network Software & Protocols**

Protocols, network simulator and software layer basics. basics.

CHAPTER 04

04

**Line Configuration & Transmission**

Point-to-point, multipoint, simplex & duplex modes. modes.

CHAPTER 05

05

**Types of Networks**

PAN, LAN, MAN, WAN, Internet / Intranet / Extranet. Extranet.

CHAPTER 06

06

**Network Architecture**

P2P vs Client-Server, Connection-Oriented vs Connectionless.

01

CHAPTER ONE · SECTION HEADER

Network Foundations

COMPUTER NETWORKS · UNIT 1

What is a computer network, why it matters, and how it evolved.

Easy

● Definition

● History

● Uses

● Applications

● Examples

1.1 CHAPTER 1 · LESSON 1

What is a Computer Network?

A computer network is a **group of two or more computers** (and other devices like printers, scanners) connected together so they can share **data, information and resources**.

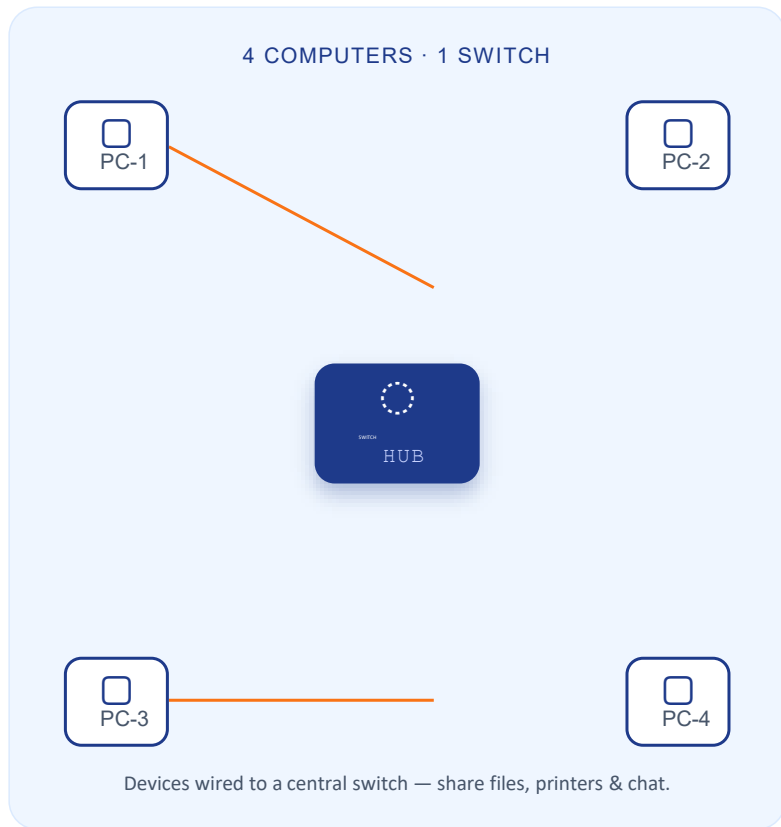
3 KEY THINGS NETWORKS ALLOW

- 1Share files & data between devices
- 2Share hardware like printers & scanners
- 3Communicate via email & chat

Q&A

Q: Define computer network.

A: A computer network is a collection of interconnected computers and devices that communicate with each other to share data, resources and information using common protocols over transmission media.



1.2 CHAPTER 1 · LESSON 2

Real-World Example: Wi-Fi at Home

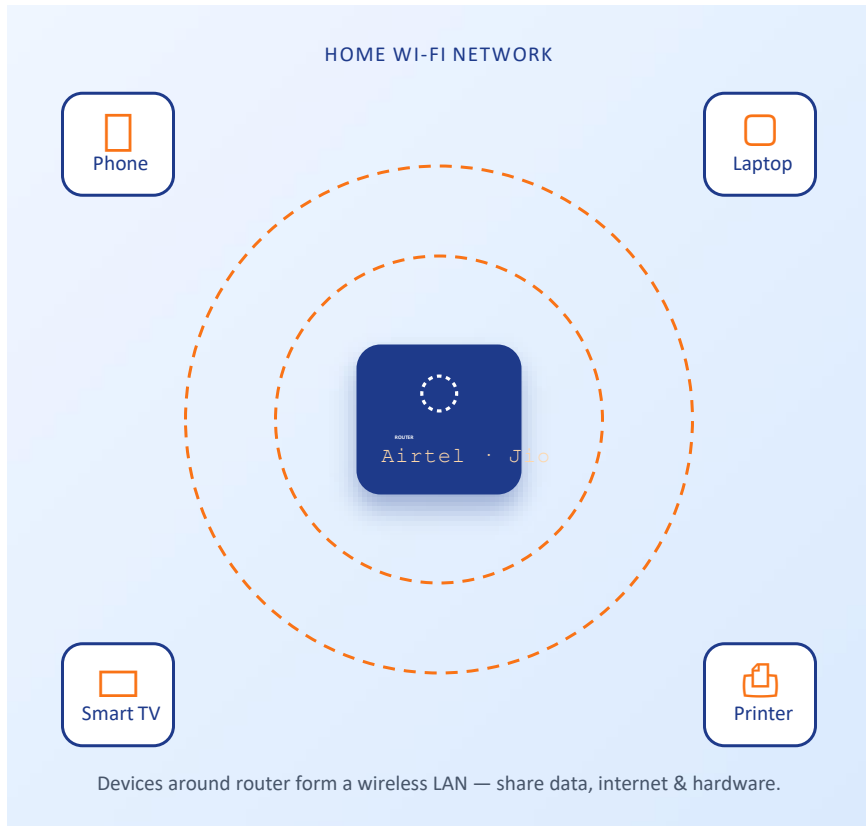
When you connect your phone, laptop, smart TV and your family's phones to the **same** the **same Wi-Fi router** at home, you have created a small computer network!

WHAT THIS HOME NETWORK LETS YOU DO

1. Stream a movie from your phone to TV
2. Print from any device to the shared printer
3. Share files between phones and laptops
4. All devices access the internet through one router

REAL EXAMPLE

Airtel/Jio Wi-Fi router at home connects 5–10 devices — phones, phones, laptops, smart TV — forming a **Local Area Network**



1.3 CHAPTER 1 · LESSON 3

Network History — How It All Started

Computer networks have come a long way. In the **1950s**, computers were huge, expensive and standalone machines. People had to **carry magnetic tapes** between computers to share data. As computers became smaller and cheaper, scientists realized they could connect them to share resources. This led to the **birth of**

1950s–60s**Mainframe Era**

One big computer, many dumb terminals connected to connected to it. No real network — just shared access.

1970s–80s**Birth of LANs**

Ethernet and ARPANET arrive. Multiple computers finally computers finally talk to each other directly.

1990s–Today**Internet Age**

Global connectivity. Wi-Fi, 4G/5G, cloud, IoT — billions billions of devices online 24/7.

ANALOGY

Think of how phones evolved — from **one landline per house** to **billions of smartphones** all connected worldwide. Computer networks evolved the same way.

1.3 CHAPTER 1 · LESSON 3 (CONT.)

Network History — Key Milestones

1957



ARPANET idea born —
USSR launches Sputnik, US
forms ARPA.

1969



ARPANET goes live —first
4 nodes connected (UCLA,
Stanford, UCSB, Utah).

1971



First email sent by Ray
Ray Tomlinson.

1983



TCP/IP becomes standard
standard —Internet is
born.

1989–91



Tim Berners-Lee invents
invents World Wide Web.
Web.

2000s–Today



Wi-Fi, 4G/ 5G, cloud,
IoT —billions of devices
devices connected.

Q&A

Q: When was ARPANET established? A: 1969.

Q: Who invented WWW? A: Tim Berners-Lee in 1989.

1.4 WHY IT MATTERS

Why Networks Matter — 5 Big Reasons

Without networks, modern life would simply stop.



01

Resource Sharing

One printer, scanner or server can serve many users —no users —no need to duplicate hardware.



02

Communication

Email, WhatsApp, Zoom all work because of networks — instant global reach.



03

Data Centralization

Store data once, access from anywhere —single source of truth across teams.



04

Speed & Efficiency

Files transfer in seconds, not hours —collaboration happens in real time.



05

Remote Work

Work from home is possible because of networks — entire teams stay productive from anywhere.



Exam Favorite

Memorize all 5 reasons —every UT/Final has at least one question on this.

REAL EXAMPLE

During COVID lockdowns in 2020, networks enabled billions of people to work, study, and shop from home.

1.5 CHAPTER 1 · LESSON 5

Uses of Computer Networks—Part 1: Communication & Sharing

Communication Uses

- Email —send/receive messages worldwide (Gmail, Outlook)
- Instant Messaging—WhatsApp, Telegram, Slack
- Video Conferencing—Zoom, Google Meet, Teams
- Voice Calls—VoIP calls via Skype, WhatsApp

Resource Sharing Uses

- Share printers/scanners across office
- Share files & folders with colleagues
- Share internet connection (one broadband, many devices)
- Share software licenses centrally

REAL EXAMPLE

In your college computer lab, **one printer is shared by all 30 PCs** through the network — that is resource sharing!

1.6 CHAPTER 1 · LESSON 6

Uses of Computer Networks — Part 2: Business & E-commerce

Business Uses

E-commerce — Amazon, Flipkart run on networks

Online Banking — UPI, net banking, ATMs

ERP Systems — manage inventory, HR, finance centrally

Cloud Computing — AWS, Azure, GCP host business apps

Customer Support — call centers, chatbots

Industrial Uses

Supply chain tracking with **RFID**

Manufacturing automation (**IoT sensors**)

Smart grid power distribution

Hospital management systems

Inventory management

Q&A

Q: List 5 business applications of computer networks.

A: E-commerce, online banking, ERP, cloud computing, customer support.

1.7 CHAPTER 1 · LESSON 7

Uses of Computer Networks—Part 3: Home & Mobile

Home Uses

- Home Wi-Fi—connect all family devices
- Smart Home—control lights, AC, fans from phone
- Streaming—Netflix, YouTube, Spotify
- Online Gaming—multiplayer games
- Smart TV—apps, casting

Mobile Uses

- Mobile Internet—4G/5G browsing
- Social Media—Instagram, Twitter
- Mobile Banking—pay bills, transfer
- Navigation—Google Maps uses network for live traffic
- App Stores—download apps, games

REAL EXAMPLE

Netflix uses networks to stream a **4K movie** to your TV from a server thousands of miles away — all in seconds.

1.8 APPLICATIONS

Major Network Applications — Where Networks Power Our World

Networks make these everyday applications possible.



01

World Wide Web

Browse websites like Google, Wikipedia — the most-used network application on Earth.



02

Electronic Mail

Send/receive emails via Gmail, Outlook — backbone of business communication.



03

Streaming Media

Netflix, YouTube, Spotify — on-demand video & audio over the network.



04

E-commerce

Amazon, Flipkart shopping — secure payments, recommendations, fast delivery.



05

Video Conferencing

Zoom, Meet, Teams — real-time audio/video calls for work & study.



06

Cloud Storage

Google Drive, Dropbox, OneDrive — files live on remote servers, sync everywhere.

ANALOGY

A network is like a **road system**. The applications (WWW, email, streaming) are like the **shops, post offices, and cinemas** that exist on those roads. Without roads, no shops. Without roads, no shops would work.

1.9 CHAPTER 1 · LESSON 9

Analogy: Computer Network = Postal System

The easiest way to understand how networks work.

Postal System		Computer Network
Letter	→	Data Packet
Sender address	→	Source IP
Receiver address	→	Destination IP
Postman (carries data between networks)	→	Router
Post office (delivers within a network)	→	Switch
Pin code	→	Subnet mask
Address format (rules of addressing)	→	Protocol
Delivery speed (data rate)	→	Bandwidth

ANALOGY

When you send a WhatsApp message, your phone (**sender**) writes a packet (**letter**) with the receiver's IP address (**address**). Routers (**postmen**) carry it across networks (**cities**) across networks (**cities**) until it reaches the receiver's phone (**destination**).

1.10 EXAM PRACTICE

Exam Q&A — Network Basics

Frequently asked 2–3 mark questions.

Q Define computer network with example.

Q1

A A computer network is a collection of interconnected devices that share data and resources. **Example:** Wi-Fi at home connecting phones, laptops, and TV.

Q When and where was ARPANET started?

Q3

A ARPANET was started in **1969** by **ARPA (US Defense Department)** connecting 4 university nodes.

Q List 4 uses of computer networks.

Q2

A Email, file sharing, printer sharing, video conferencing.

Q Name the inventor of WWW and year.

Q4

A **Tim Berners-Lee** invented World Wide Web in **1989**.

TIP Always give an **example** with the definition — earns extra mark!

CHAPTER TWO · SECTION HEADER

02

Network Hardware & Servers

COMPUTER NETWORKS · UNIT 1

The physical machines that make networks work.

Easy

● Workstations

● Servers

● Types of Servers

● Hardware Overview

2.1 CHAPTER 2 · LESSON 1

Network Hardware — The Building Blocks

Physical devices that make up a computer network.



01

Workstation / Node

Any computer connected to the network — what users actually use.

USER DEVICE



02

Server

Powerful computer that serves other devices on the network.
network.

SERVICE PROVIDER



03

Router

Connects **different networks** together (e.g. home network to Internet).
Internet).

INTER-NETWORK



04

Switch / Hub

Connects devices **within the same network**; forwards frames by MAC.
MAC.

INTRA-NETWORK



05

NIC

Network Interface Card — connects a device to the network (wired or wireless).

ADAPTER



06

Transmission Media

Cables (wired — UTP, fiber) or Wi-Fi (wireless) — the physical path.

MEDIUM

REAL EXAMPLE

Your home Wi-Fi setup has: a **router** (from Airtel/Jio), **NIC** inside your laptop, and **wireless media** (Wi-Fi signal).

2.2 CHAPTER 2 · LESSON 2

Nodes & Workstations — The Network Users

A **node** is any device connected to a network — it can be a computer, laptop, phone, printer, scanner, or even a smart TV. A **workstation** is a special kind of node — it is a powerful computer used by a person to do work, and it requests services from servers.

- ✓Every device on a network is a node
- ✓Workstations are user computers that REQUEST services
- ✓Servers PROVIDE services to workstations
- ✓Workstations are also called 'clients' in client-server model

Q&A

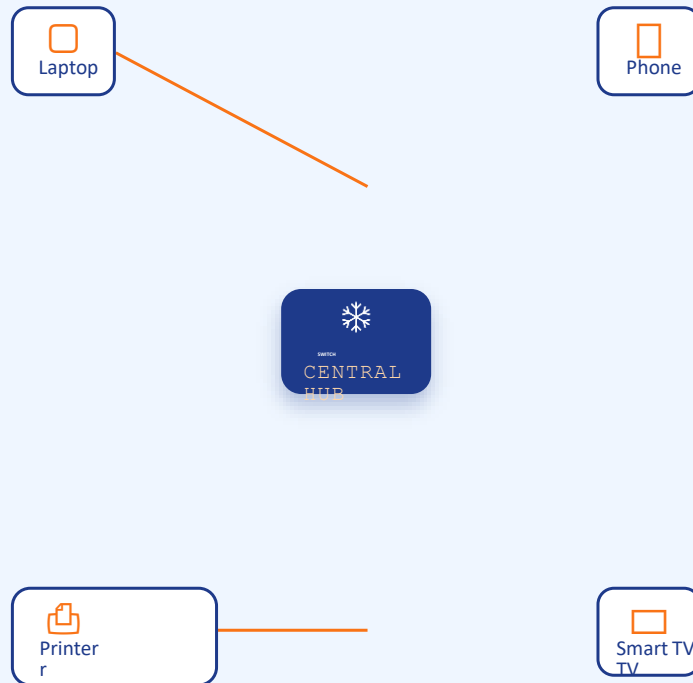
Q: What is a node?

A: Any device connected to a network that can send, receive or forward information.

Q: Differentiate node and workstation.

A: All workstations are nodes, but not all nodes are workstations (e.g. printer is a node but not workstation).

DIFFERENT NODES · ONE NETWORK



Laptop = workstation · Printer/TV/Phone = nodes (not workstations)

2.3 CHAPTER 2 · LESSON 3

What is a Server?

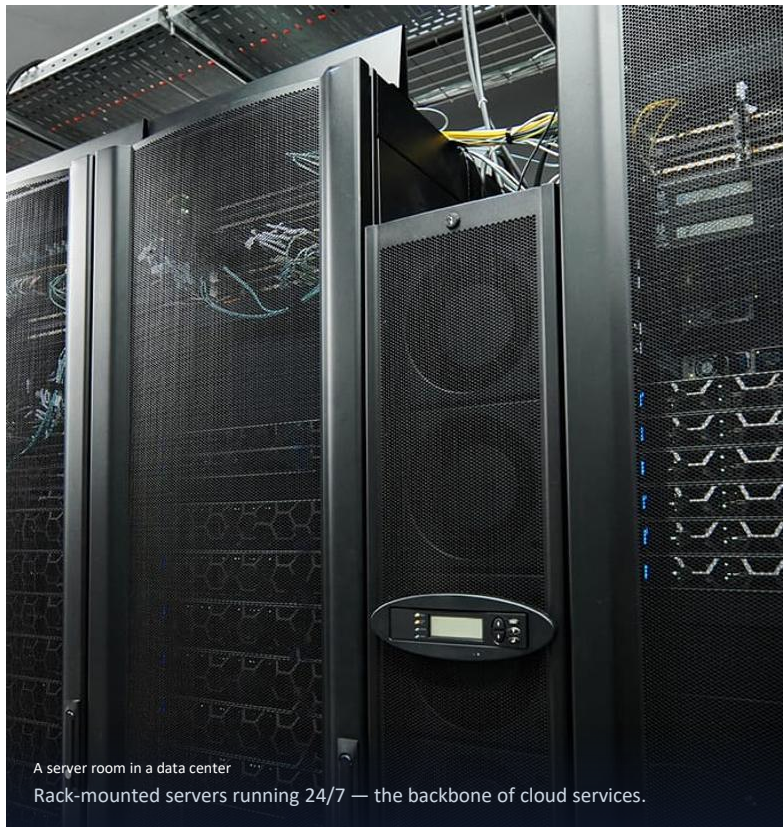
A **server** is a powerful computer designed to **provide services, data, or resources** to other computers (called clients) on a network. Servers run 24/7 and are built to handle many requests at requests at once. They are usually more powerful, have more memory, bigger storage, and faster faster processors than normal computers.

SERVER CHARACTERISTICS

- ⚡ Always switched ON (24/7 uptime)
- 🖨 High processing power & memory
- ⬆ Runs special server software (Windows Server, Linux)
- #Has static IP address (doesn't change)
- ❄ Stored in cool, secure server rooms

REAL EXAMPLE

When you open **google.com**, your request goes to **Google's server** (a powerful (a powerful computer in a data center) which sends back the search results.



A server room in a data center

Rack-mounted servers running 24/7 — the backbone of cloud services.

2.4 CHAPTER 2 · LESSON 4

Analogy: Server = Waiter in a Restaurant

Easy way to remember the client-server concept.

Restaurant		Computer Network
👤 Waiter (serves food)	→	Server (serves data / files / services)
😊 Customer (orders food)	→	Client (requests data / files)
🍳 Kitchen (food prepared)	→	Server storage (data stored)
📋 Menu (list of dishes)	→	Protocol (list of services)
📋 Order placed	→	Request sent
🍽️ Food delivered	→	Response received
👥 Multiple customers served	→	Multiple clients served simultaneously

ANALOGY

In **McDonalds**, the waiter (server) takes orders from many customers (clients) at once, gets food from kitchen (storage), and delivers to each table. A computer server works exactly the same way — many clients send requests, server processes all of them and sends responses back.

2.5 CHAPTER 2 · LESSON 5

Types of Servers — Part 1

The most common servers you use daily.



01

File Server

Stores and manages files centrally. Users can save, open save, open and share files from one location.

EXAMPLE

Office shared drive (**Z: drive**) — every employee's employee's "home" folder.



02

Print Server

Manages print requests from multiple computers to one computers to one or more printers. Queues print jobs jobs so they don't mix up.

EXAMPLE

College lab printer shared by **30 PCs** — prints don't collide.



03

Web Server

Stores website files and serves them to browsers when when users visit a website.

EXAMPLE

Apache, Nginx, IIS serve websites like google.com, youtube.com.

REAL EXAMPLE

When you visit **youtube.com**, YouTube's **web server** sends video files to your browser.

2.6 CHAPTER 2 · LESSON 6

Types of Servers — Part 2

Servers that handle data, communication and apps.



Mail Server

04

Stores, sends and receives emails. Uses protocols like SMTP (send), POP3/IMAP (receive).

⇄ SMTP · POP3 · IMAP

EXAMPLE

Gmail's mail server handles millions of emails per minute.



Database Server

05

Stores and manages databases. Handles queries from applications.

} SQL Queries · ACID

EXAMPLE

MySQL, Oracle, SQL Server store data for banks, e-commerce sites.



Application Server

06

Runs business applications and provides business logic to clients.

◆ Java EE · Middleware

EXAMPLE

WebSphere, Tomcat run Java web apps for banking, ERP.

REAL EXAMPLE

When you check account balance in banking app, request goes to **application server**, which asks **database server** for your balance, and sends result back to your phone.

2.7 CHAPTER 2 · LESSON 7

Types of Servers — Part 3

Specialized servers for security, naming and more.



07

SECURITY · INTERMEDIARY

Proxy Server

Acts as middleman between client and internet. Hides client identity, caches content, blocks sites.

EXAMPLE

College proxy blocks social media during lectures.



08

NAMING · ADDRESS RESOLUTION
RESOLUTION

DNS Server

Translates website names (google.com) to IP addresses (142.250.190.46). Without DNS, you'd have to memorize numbers.

EXAMPLE

Google DNS at **8.8.8.8**.



09

REAL-TIME · MULTIPLAYER

Game Server

Hosts multiplayer online games. Handles player positions, scores, chat.

EXAMPLE

PUBG, Free Fire, BGMI servers.

Q&A

Q: What is proxy server?

A: A server that acts as intermediary between client and other servers, providing security, caching and access control.

CHAPTER 2 · SERVER TYPES

All Server Types at a Glance

Server Type	Purpose	Example	Protocol Used
File Server	Stores & shares files	Office shared drive	SMB, FTP
Print Server	Manages printers	Office printer queue	IPP
Web Server	Serves websites	Apache, Nginx	HTTP, HTTPS
Mail Server	Sends & receives emails	Gmail, Exchange	SMTP, POP3, IMAP
Database Server	Stores databases	MySQL, Oracle	SQL
Application Server	Runs applications	Tomcat, WebSphere	HTTP, RMI
Proxy Server	Middleman for security	Squid, CCProxy	HTTP, SOCKS
DNS Server	Name to IP translation	Google DNS 8.8.8.8	DNS
Game Server	Hosts online games	PUBG, BGMI servers	UDP

EXAM TIP Tip: Memorize this table — 1 question is sure to come from server types.

✦ EXAM PRACTICE

Exam Q&A — Servers & Hardware

1 What is a server? Differentiate from workstation.

Server is a powerful computer that provides services to clients on a network.

Workstation is a user's computer that requests services.

2 List 6 types of servers with their use.

File (storage), **Print** (printing), **Web** (websites), **Mail** (emails), **Database** (data), **Application** (apps).

3 What is a proxy server and its uses?

Server acting as **middleman** for security, caching, access control and anonymity.

4 Which server translates google.com to IP?

DNS Server (Domain Name System) translates domain names to IP addresses.

EXAM TIP

Tip: For server questions, always mention at least 1 example — earns full marks.

CHAPTER

03

📄 Slides 27–35

Network Software & Protocols

The rules and tools that make networks communicate.

Network Software

Protocols

Common Protocols

Network Simulator

CHAPTER 3 · LESSON 1

Network Software — The Brain Behind Networks

Software that controls and manages network operations.



Protocols

Rules for communication (HTTP, TCP, IP).



Network OS

Software that runs servers (Windows Server, Linux).



Network Simulator

Software to test networks (Cisco Packet Tracer).



Network Security Software

Firewalls, antivirus.



Network Management Software

Monitors network health.



Client Software

Browsers, email clients, FTP clients.

REAL EXAMPLE

Cisco Packet Tracer is network simulator software used in this course — you will use it in lab assignments.

CHAPTER 3 · LESSON 2

What is a Protocol?

A **protocol** is a set of rules that defines how data is transmitted between two devices on a network. Just like humans need a common language to talk, computers need a common protocol to communicate. Without protocols, two computers cannot understand each other — even if they are physically connected.

3 KEY THINGS PROTOCOLS DEFINE

1. **Format of messages** — how data is packaged
2. **Order of messages** — who sends what first
3. **Error handling** — what to do if data is lost

ANALOGY

Imagine two people — one speaks only Hindi, other speaks only French. They cannot communicate. But if both know English, they can talk. **English here is the here is the protocol** — a common language. Similarly, two computers need a common protocol to talk.

Q & A

Q: Define protocol.

A: A protocol is a set of rules and conventions that govern data communication between networked devices, including message format, order, and error handling.



Two devices exchanging data using a shared protocol.

CHAPTER 3 · LESSON 3

Common Protocols — Web & File Transfer



HTTP

HYPERTEXT TRANSFER PROTOCOL

PORT 80

Used to transfer web pages from server to browser. Used by every website.

Data sent in **plain text (insecure)**.

Example: visiting `http://example.com`



HTTPS

HTTP SECURE

PORT 443

Same as HTTP but with SSL/TLS encryption. Secures passwords, payments, banking.

Encrypted — **safe for sensitive data**.

Example: `https://www.onlinesbi.com`



FTP

FILE TRANSFER PROTOCOL

PORT 21

Used to upload/download files between client and server. Used by web developers to upload website files.

Example: uploading website to hosting server using FileZilla.

REAL EXAMPLE

The lock icon you see in browser address bar means **HTTPS** is protecting your data with encryption.

CHAPTER 3 · LESSON 4

Email Protocols — Sending & Receiving Mails



SMTP

SIMPLE MAIL TRANSFER PROTOCOL

PORT 25 → SEND

Used to **SEND** emails from client to mail server, and between mail servers. Always used for sending.

Example: when you click "Send" in Gmail, SMTP carries your email.



POP3

POST OFFICE PROTOCOL V3

PORT 110 ↓ RECEIVE · DELETES

Used to **RECEIVE** emails. Downloads email from server to client, then deletes from server. Good for single device.

Example: Outlook configured with POP3.



IMAP

INTERNET MESSAGE ACCESS PROTOCOL

PORT 143 ↔ SYNC · KEEPS

Used to **RECEIVE** emails. Syncs emails between client and server (does not delete). Best for multiple devices.

Example: Gmail app on phone + laptop shows same emails.

EXAM Q&A

Q: Which protocol sends email? **A:** SMTP. **Q:** Which protocol is used to receive email on multiple devices? **A:** IMAP (syncs, doesn't delete).

CHAPTER 3 · LESSON 5

Core Protocols — TCP, UDP, IP

The backbone protocols of the internet.



TRANSMISSION CONTROL
PROTOCOL

RELIABLE · CONNECTION-ORIENTED

Connection-oriented, reliable. Guarantees data delivery in order. Resends lost packets. Slower.

Used for: web browsing, email, file transfer.

Example: loading a webpage.



USER DATAGRAM
PROTOCOL

FAST · CONNECTIONLESS

Connectionless, fast. No guarantee of delivery. No resend. Faster than TCP.

Used for: video calls, gaming, live streaming.

Example: WhatsApp video call, PUBG.



INTERNET
PROTOCOL

ADDRESSING · ROUTING

Handles addressing and routing. Every device has a unique IP. Two versions: IPv4 (32-bit) and IPv6 (128-bit).

Example: 192.168.1.1 (IPv4)

EXAM Q&A

Q: Difference between TCP and UDP? **A:** TCP is reliable, connection-oriented, slower. UDP is unreliable, connectionless, faster. TCP used for web/email, UDP for video/gaming.

CHAPTER 3 · LESSON 6

Analogy: Protocols = Languages Humans Speak

Why different protocols exist for different needs.

Human World		Network World	Why it fits
English (global language)	→	IP	Universal addressing
Phone call (continuous conversation)	→	TCP	Reliable, ordered
SMS text (quick one-way)	→	UDP	Fast, no guarantee
Post office rules	→	HTTP/HTTPS	Web page transfer rules
Courier service	→	FTP	File transfer
Speed post vs ordinary post	→	TCP vs UDP	Speed vs reliability
Different country, different language	→	Different protocol	Different protocol for different app

ANALOGY

Just like you use Hindi to chat with friends, English for office emails, and sign language with someone deaf — computers use **TCP for websites**, **UDP for video calls**, and **FTP for file uploads**. Different jobs need different protocols!

CHAPTER 3 · LESSON 7

Network Simulator — Test Before You Build

A **network simulator** is a software program that mimics (imitates) how a real computer network behaves. It lets you design, build and test networks virtually — without buying any real hardware. You can add routers, switches, PCs, cables, configure IPs and see how data flows — all on your computer screen.

WHY SIMULATORS ARE USED

1. **Save cost** — no real hardware needed.
2. **Safe** — mistakes don't damage real equipment.
3. **Learn fast** — test topologies before deployment.
4. **Visual** — see packets moving in real time.

REAL EXAMPLE

In this course's lab, you will use **Cisco Packet Tracer** to design 6 topologies — bus, ring, bus, ring, star, mesh, tree, hybrid.

Q & A

Q: What is network simulator?

A: Software that imitates real network behavior, allowing users to design, test and analyze networks without physical hardware.

**POPULAR SIMULATORS**

Cisco Packet Tracer



Course tool



GNS3

Pr

o



NS2

Resea
rch

NS3

Resea
rch

OMNeT++

Acade
mic

Cisco Packet Tracer is the simulator for this course.

CHAPTER 3 · LESSON 8

Cisco Packet Tracer — Your Lab Tool

Cisco Packet Tracer is a network simulator developed by Cisco Systems. It is the most popular simulator used by students and networking professionals to learn networking concepts, design topologies and practice router/switch configuration — without needing real Cisco hardware.



01

Drag & Drop Devices

Add routers, switches, PCs, servers from device library to library to workspace.



02

Configure Devices

Click any device to configure IP, routing, interfaces just like real Cisco IOS.



03

Simulate Packets

Run simulation mode to see packets traveling between devices step by step.

REAL EXAMPLE

In your lab assignments 2–7, you will build **bus, ring, star, mesh, tree and hybrid** topologies in Packet Tracer.

✦ EXAM PRACTICE

Exam Q&A — Protocols & Simulator

1 Define protocol with example.

Set of rules governing communication between devices. **Example:** HTTP for web, SMTP for email.

2 Differentiate TCP and UDP.

TCP is connection-oriented, reliable, slower — used for web/email. **UDP** is connectionless, unreliable, faster — used for video calls/gaming.

3 What is HTTP and HTTPS?

HTTP (port 80) transfers web pages in plain text. **HTTPS** (port 443) does same but with same but with SSL/TLS encryption for security.

4 What is a network simulator? Name one.

Software that mimics real network behavior for testing. **Example:** Cisco Packet Tracer.

EXAM TIP

For protocol questions, always write port numbers:

80 HTTP

443 HTTPS

25 SMTP

21 FTP

CHAPTER

04

Slides 37–44

Line Configuration & Transmission Transmission Modes

How devices connect and which direction data flows.

Point-to-Point

Multipoint

Simplex

Half-Duplex

Full-Duplex

CHAPTER 4 · LESSON 1

Line Configuration — Point-to-Point

In **point-to-point line configuration**, a dedicated link is established between exactly two devices. The entire capacity of the link is reserved for communication between these two devices only. No other device shares this link.

KEY POINTS

- Only 2 devices connected
- Dedicated link — no sharing
- Full bandwidth available
- Simple to set up
- Example: TV remote to TV, mouse to computer

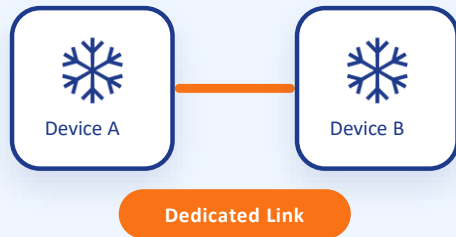
REAL EXAMPLE

When you connect your phone to your laptop with a USB cable for file transfer, transfer, that's a point-to-point connection — only your phone and laptop are using that link.

Q & A

Q: What is point-to-point configuration? **A:** A dedicated link between two devices where devices where the full channel capacity is reserved for them only.

POINT-TO-POINT LINK



Full bandwidth reserved for these two devices only.

CHAPTER 4 · LESSON 2

Line Configuration — Multipoint

In **multipoint line configuration**, a single shared link is used by multiple devices. All devices share the same communication channel, so they must take turns or follow rules to avoid data collision. The bandwidth of the link is shared among all connected devices.

KEY POINTS

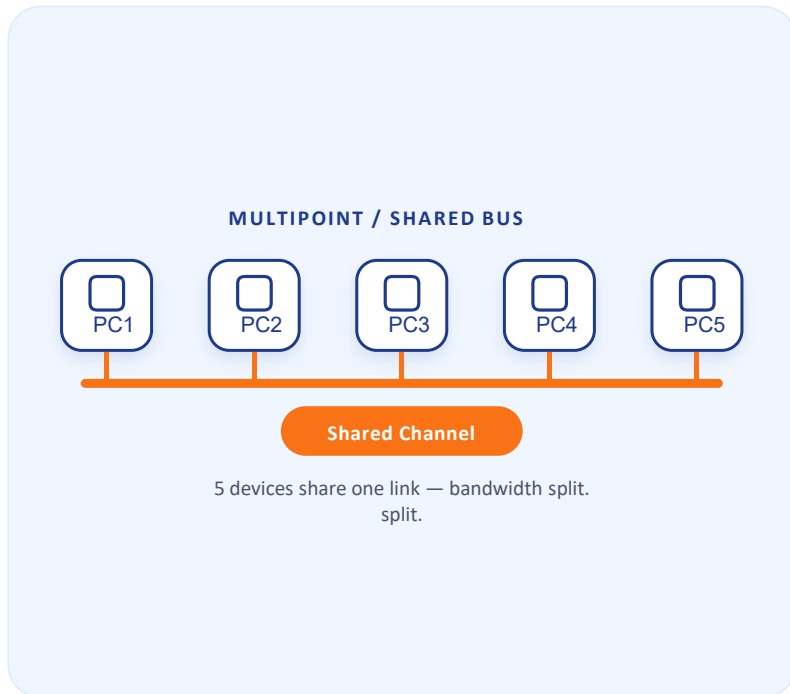
- More than 2 devices share one link
- Bandwidth is shared (less per device)
- Need rules to avoid collision
- Cheaper (less cabling)
- Example: bus topology, Wi-Fi network

REAL EXAMPLE

In old computer labs with bus topology, all PCs were connected to a single coaxial cable — that's multipoint configuration. Today, Wi-Fi works the same way — all phones share the wireless channel.

Q & A

Q: Differentiate point-to-point and multipoint. **A:** Point-to-point: dedicated link between 2 devices, full bandwidth. Multipoint: shared link between many devices, bandwidth shared.



CHAPTER

05

Slides 46–56

Types of Networks

PAN, LAN, MAN, WAN — and the Internet family.

PAN

LAN

MAN

WAN

Internet

Intranet

Extranet

05·1 CHAPTER 5 - LESSON 1

How Networks Are Classified — By Size

4 main types based on the geographical area covered.

RANGE ~10 m

PAN — Personal Area Network

Smallest network. Covers 1 person, ~10 meters. Connects personal devices.

Example: Bluetooth earphones to phone.

RANGE ~1 km

LAN — Local Area Network

Covers a building or campus. Up to ~1 km. Owned by one organization.

Example: College computer lab, office Wi-Fi.

RANGE ~50 km

MAN — Metropolitan Area Network

Covers a city. Up to ~50 km. Connects multiple LANs in the city.

Example: Cable TV network, city-wide Wi-Fi.

RANGE UNLIMITED

WAN — Wide Area Network

Largest. Covers countries or continents. Thousands of kilometers.

Example: Internet, ATMs across India.

EXAM Q&A

Q: Classify networks by size. **A:** PAN (10 m, personal), LAN (1 km, building), MAN (50 km, city), WAN (country/continent).

05·2 CHAPTER 5 - LESSON 2

PAN — Personal Area Network

A **Personal Area Network (PAN)** is the smallest type of network. It connects devices within the personal space of a single person — usually within 10 meters. PAN is built around one person and connects their personal devices like phone, laptop, headphones, smartwatch, etc.

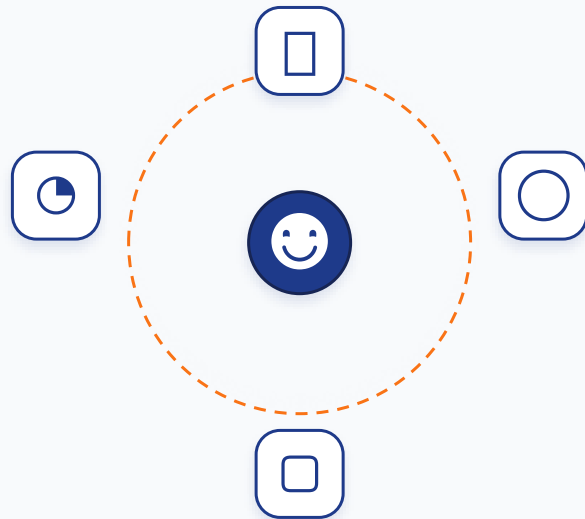
- **Range:** ~10 meters
- Connects personal devices
- Wired (USB) or wireless (Bluetooth)
- Owned by one person
- Wireless PAN called **WPAN**

REAL EXAMPLES

(1) Bluetooth earphones connected to phone — that is a PAN · (2) Smartwatch paired with phone · (3) Wireless mouse and keyboard · (4) Phone connected to car audio via Bluetooth.

EXAM Q&A

Q: What is PAN? **A:** Personal Area Network — network within 10 m connecting one connecting one person's devices. Example: Bluetooth earphones + phone.

PAN · 10 m radius

One person · personal devices · Bluetooth/USB

05·3 CHAPTER 5 - LESSON 3

LAN — Local Area Network

A **Local Area Network (LAN)** connects computers and devices within a small geographical area — like a area — like a building, office, school, or college campus. LAN is the most common type of network. It is network. It is usually owned, set up and maintained by a single organization.

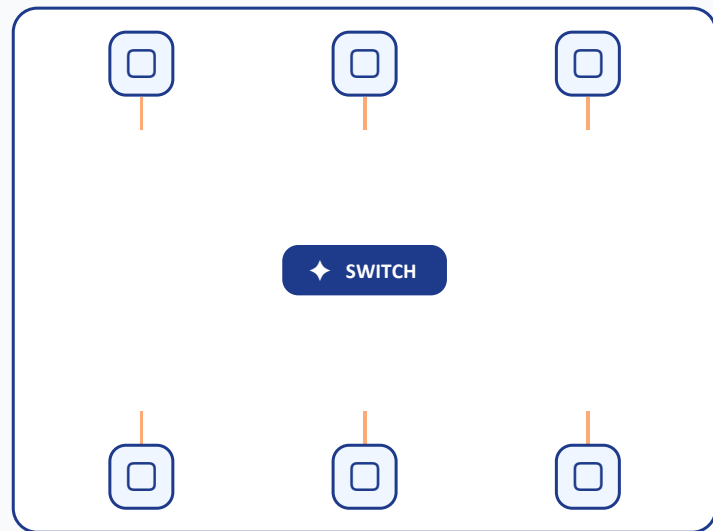
- **Range:** up to ~1 km (building/campus)
- High speed: 100 Mbps to 10 Gbps
- Low error rate
- Privately owned
- Wired (Ethernet) or wireless (Wi-Fi = WLAN)

REAL EXAMPLES

(1) College computer lab with 30 PCs connected by Ethernet · (2) Office Wi-Fi connecting employees' laptops · (3) Cyber cafe network · (4) Home Wi-Fi connecting family devices.

EXAM Q&A

Q: What is LAN? Give 2 examples. **A:** Local Area Network — connects within a building/campus up to ~1 km. Examples: college computer lab, Fi.

LAN · 1 BUILDING · ~1 km

Multiple PCs · 1 switch · 1 building

05·4 CHAPTER 5 - LESSON 4

MAN — Metropolitan Area Network

A **Metropolitan Area Network (MAN)** connects computers across a city or a metropolitan area. It is area. It is bigger than a LAN but smaller than a WAN. A MAN typically connects multiple LANs together — LANs together — for example, all branches of a bank in one city are connected via a MAN.

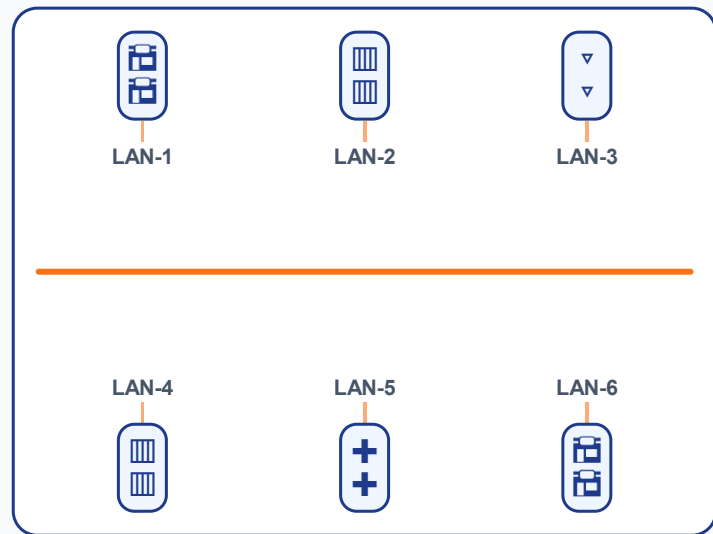
- **Range:** ~5 to 50 km (city-wide)
- **Speed:** moderate (10–100 Mbps)
- **Connects multiple LANs**
- **Owned by single org or service provider**
- **Used by city-wide businesses**

REAL EXAMPLES

(1) Cable TV network distributing channels to homes in a city · (2) All SBI branches in Mumbai connected via MAN · (3) City-wide free Wi-Fi like in Pune or Bangalore · (4) University campuses spread across city.

EXAM Q&A

Q: What is MAN? **A:** Metropolitan Area Network — connects networks within a city, within a city, range ~5–50 km. Example: Cable TV network in a city.

MAN · 1 CITY · ~5–50 km

Multiple LANs · 1 MAN backbone · 1 city

05·5 CHAPTER 5 - LESSON 5

WAN — Wide Area Network

A **Wide Area Network (WAN)** is the largest type of network. It connects computers and smaller networks smaller networks (LANs and MANs) across large geographical areas — states, countries, or even even continents. WANs use public or private communication links like telephone lines, satellites, and satellites, and undersea cables.

- **Range:** countries/continents (unlimited)
- Lower speed than LAN (due to distance)
- High error rate
- Owned by multiple service providers
- Most expensive to set up

REAL EXAMPLES

(1) The INTERNET is the biggest WAN in the world! · (2) ATM network of all banks across India · (3) Indian Railways reservation network across all stations · (4) Multinational company offices connected worldwide.

EXAM Q&A

Q: What is WAN? Give 2 examples. **A:** Wide Area Network — spans countries/continents. Examples: Internet, ATM banking network across

WAN · COUNTRIES & CONTINENTS

WAN connects the world

05·6 CHAPTER 5 - LESSON 6

LAN vs MAN vs WAN — Full Comparison

Feature	LAN	MAN	WAN
Full Form	Local Area Network	Metropolitan Area Network	Wide Area Network
Coverage	Building/Campus (~ 1 km)	City (~ 5–50 km)	Country/Continent (unlimited)
Speed	Very High (100 Mbps – 10 Gbps)	Moderate (10–100 Mbps)	Low (56 Kbps – 1 Gbps)
Ownership	One organization	One or more orgs	Multiple service providers
Cost	Low	Medium	High
Error Rate	Very Low	Medium	High
Setup	Easy	Difficult	Very Difficult
Examples	College lab, Office Wi-Fi	Cable TV, City bank network	Internet, ATM network

EXAM Q&A

Q: Differentiate LAN, MAN, WAN (any 5 points). **A:** See table above — cover range, speed, ownership, cost, example.

05.7 CHAPTER 5 - LESSON 7

Internet — The Global Network

The **Internet** is the world's largest Wide Area Network (WAN). It is a global network of networks — connecting billions of computers, smartphones, servers and other devices across the world. The Internet uses TCP/IP protocol suite to link devices globally. Anyone with an internet connection can access information from anywhere on the planet.

- World's largest network
- Network of networks (connects LANs, MANs, WANs)
- Uses TCP/IP protocols
- Public network — anyone can connect
- Birth: 1983 (TCP/IP became standard)

REAL EXAMPLES

You use Internet to browse Google, watch YouTube, send WhatsApp, post WhatsApp, post Instagram, attend Zoom classes, shop on Amazon, do UPI do UPI payments — all of this runs on the Internet.

EXAM Q&A

Q: What is Internet? **A:** The Internet is a global WAN connecting billions of devices of devices worldwide using TCP/IP protocols. It is a network of networks.



The Internet connects the world

05·8 CHAPTER 5 - LESSON 8

Intranet — Private Company Network

An **Intranet** is a PRIVATE network that belongs to an organization and is accessible only by that organization's employees or members. It uses the SAME technology as the Internet (TCP/IP, browsers) (TCP/IP, browsers) but access is restricted. Think of it as the Internet just for one company — outsiders cannot enter.

- Private network of an organization
- Only employees/members can access
- Uses Internet technology (TCP/IP, browsers)
- Behind firewalls for security
- Used to share internal info

REAL EXAMPLES

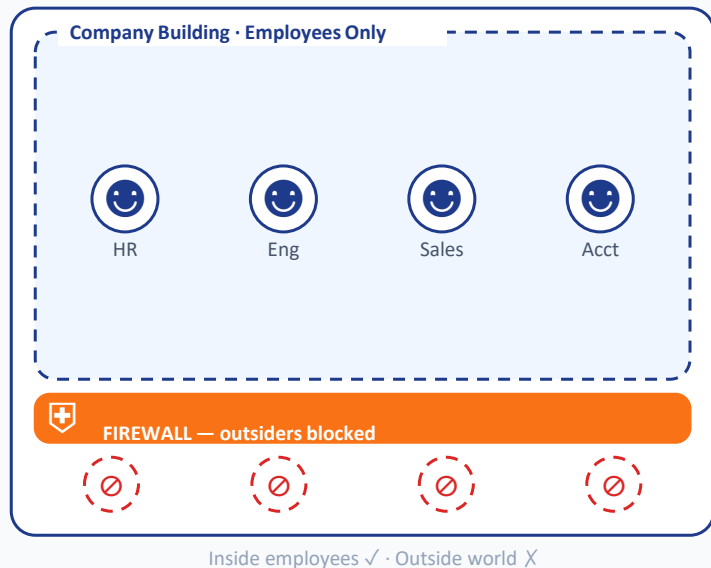
(1) College ERP portal (attendance, marks) — only students/staff log in · (2) Company HR portal for leave applications · (3) Bank's internal employee portal · (4) Indian Railways' internal staff website.

ANALOGY

Intranet is like a company's private notice board — only employees can read it, read it, outsiders cannot. Same language (English) as public boards, but access is restricted.

EXAM Q&A

Q: What is Intranet? **A:** A private network using Internet technology, accessible only to authorized members of an organization.

INTRANET · ONE ORGANIZATION

05·9 CHAPTER 5 - LESSON 9

Extranet — Extended Private Network

An **Extranet** is an extended version of an Intranet. It allows controlled access to the organization's private organization's private network to OUTSIDE people — like customers, suppliers, partners, vendors — who vendors — who need to work with the organization. The outside access is limited and granted only to granted only to authorized external users.

- Extension of an Intranet
- Allows controlled outside access
- For customers, suppliers, partners
- More open than Intranet, less open than Internet
- Requires login and authentication

REAL EXAMPLES

(1) Flipkart seller portal — sellers login to manage products · (2) Dell's portal for suppliers to check orders · (3) College online admission portal open to students · (4) Vendor portal of Maruti Suzuki for parts suppliers.

ANALOGY

If Intranet is a company's private office, Extranet is when the company allows selected outsiders (delivery agents, clients) to enter specific rooms — not all — not all rooms. Internet, in contrast, is the public road outside.

EXAM Q&A

Q: What is Extranet? **A:** Extension of Intranet that allows controlled access to

EXTRANET · ORG + AUTHORIZED PARTNERS



LOGIN & AUTHENTICATION GATE

Intranet (employees) + authorized external partners

05:10 CHAPTER 5 - LESSON 10

Internet vs Intranet vs Extranet

Feature	Internet	Intranet	Extranet
Access	Public (anyone)	Private (employees only)	Private + authorized outsiders
Network Type	Global WAN	Private LAN/WAN	Extension of Intranet
Size	Largest (global)	One organization	One org + partners
Security	Low (open)	High (firewall)	Medium-high (login)
Ownership	No single owner	One organization	One organization
Example	Google, YouTube	College ERP portal	Flipkart seller portal
Users	Billions of people	Employees/students	Employees + suppliers/customers
Cost to use	Free (just need ISP)	Free for employees	Free but login needed

EXAM Q&A

Q: Differentiate Internet, Intranet, Extranet. **A:** Internet = public global; Intranet = private org-only; Extranet = private + authorized outsiders.

05·11 CHAPTER 5 - LESSON 11

Exam Q&A — Types of Networks

CLASSIFY BY SIZE

Q1

Classify networks by size with range.

A: PAN (10 m, personal), LAN (1 km, building), MAN (50 km, city), WAN (country, continent). continent).

LAN vs MAN vs WAN

Q2

Differentiate LAN, MAN, WAN (3 points).

A: **Range:** LAN<1 km, MAN<50 km, WAN unlimited. **Speed:** LAN highest, WAN lowest. **Ownership:** LAN one org, WAN multiple providers.

DEFINITION

Q3

What is Internet?

A: Global WAN connecting billions of devices using TCP/IP. Network of networks.

INTRANET vs EXTRANET

Q4

Differentiate Intranet and Extranet.

A: Intranet = private network for employees only. Extranet = extends Intranet to authorized outside partners (suppliers, customers).

TIP

Always memorize the 4 network ranges — PAN (10 m), LAN (1 km), MAN (50 km), WAN (unlimited).

CHAPTER 06 NETWORK ARCHITECTURE & SERVICES

06

Network Architecture & Services

P2P vs Client-Server and Connection-Oriented vs Connectionless.

Easy

P2P

Client-Server

Connection-Oriented

Connectionless

06 · 1 CHAPTER 6 - LESSON 1

Peer-to-Peer Network (P2P)

In a **Peer-to-Peer (P2P) network**, all computers (called peers) have EQUAL status. There is NO central server. Every computer can act as both a client AND a server — it can request files from others AND share its own files with others. All peers are equal in responsibility and capability.

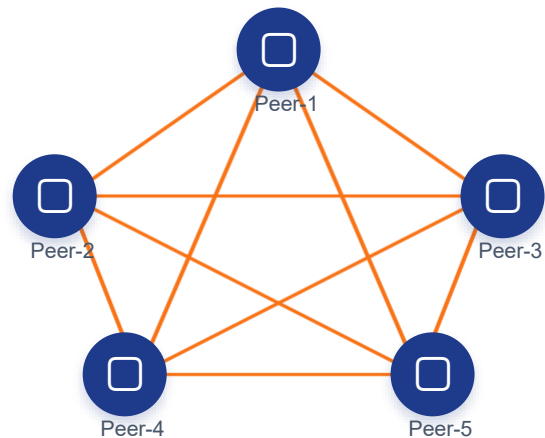
- No central server — all peers equal
- Each peer = client + server
- Easy and cheap to set up
- Works well for small networks (up to 10 computers)
- Hard to manage as network grows
- Less secure

REAL EXAMPLES

(1) BitTorrent — file sharing where everyone uploads and downloads · (2) Skype (original) — calls made directly between users · (3) Sharing files between 2 laptops via Wi-Fi Direct · (4) Bitcoin network — all nodes are peers.

EXAM Q&A

Q: What is P2P network? **A:** Network where all computers have equal central server, each peer acts as both client and server.

P2P · MESH OF EQUAL PEERS

5 peers · all equal · all connected to each other

06·2 CHAPTER 6 - LESSON 2

Client-Server Network

In a **Client-Server network**, there is ONE central powerful computer called the **SERVER** that provides services, and many **CLIENT** computers that request those services. The server is the boss — it controls everything. Clients send requests, server responds. This is the most common architecture in modern networks.

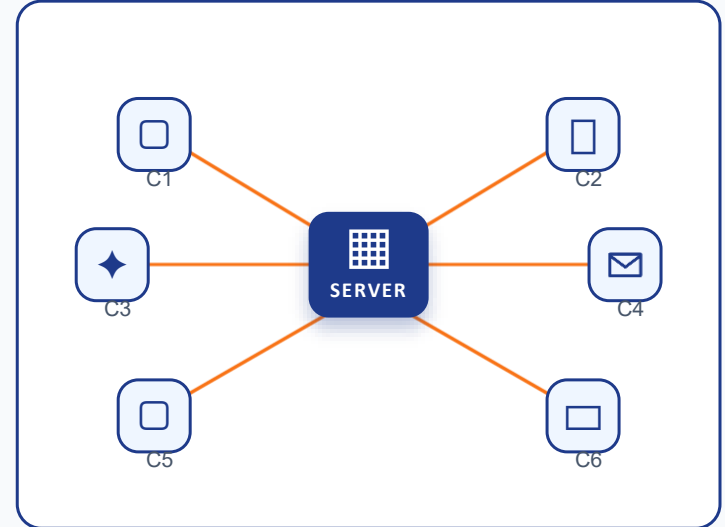
- One central server, many clients
- Server provides services
- Clients only request — they don't serve
- Easy to manage and secure
- Scales well to large networks
- If server fails, entire network stops

REAL EXAMPLES

(1) WhatsApp — your phone (client) connects to WhatsApp server ·
(2) ATM — ATM machine (client) requests money from bank server · (3)
Website — browser (client) requests pages from web server · (4)
Email — Outlook (client) fetches mail from mail server.

EXAM Q&A

Q: What is client-server network? **A:** Network with one central server that provides services to multiple client computers that request them.

CLIENT-SERVER · 1 BOSS, MANY WORKERS

1 server · 6 clients · all requests flow to server

06·3 CHAPTER 6 - LESSON 3

Peer-to-Peer vs Client-Server Network

Feature	Peer-to-Peer (P2P)	Client-Server
Architecture	All peers equal	One server, many clients
Server	No dedicated server	Central dedicated server
Cost	Low (no special hardware)	High (need powerful server)
Management	Difficult (each peer manages self)	Easy (centralized)
Security	Low (decentralized)	High (centralized control)
Scalability	Limited (up to ~10 peers)	High (thousands of clients)
Failure Impact	One peer down = small impact	Server down = entire network down
Best For	Small home/file sharing	Large org, businesses
Examples	BitTorrent, Skype, Bitcoin	WhatsApp, Google, Banking

EXAM Q&A

Q: Differentiate P2P and Client-Server (5 points). **A:** See table — architecture, server, cost, security, scalability, examples.

06·4 CHAPTER 6 - LESSON 4

Connection-Oriented Service

In **connection-oriented service**, a dedicated connection is **FIRST** established between sender and receiver **BEFORE** any data is sent. After data transfer is complete, the connection is properly closed. It is like making a phone call — you first dial, wait for the other person to pick up, then talk, then hang up.

- **CONNECTION SETUP** — sender & receiver establish connection (handshake)
- **DATA TRANSFER** — data is sent reliably over the connection
- **CONNECTION RELEASE** — connection is properly closed after transfer

REAL EXAMPLES

(1) TCP protocol — used for web browsing, email, file transfer · (2) Telephone call — dial, talk, hang up · (3) ATM withdrawal — insert enter PIN, transact, take card.

EXAM Q&A

Q: What is connection-oriented service? **A:** Service where a connection is established before data transfer and released after. Reliable but slower. Example: TCP.

3 PHASES · TCP / PHONE CALL

1

**Connection Setup**

Handshake — establish connection



2

**Data Transfer**

Reliable data flow over connection



3

**Connection Release**

Properly close the connection

Setup → Transfer → Release

06·5 CHAPTER 6 - LESSON 5

Connectionless Service

In **connectionless service**, data is sent WITHOUT establishing any connection first. The sender simply sends packets — and hopes they reach the destination. There is no setup, no acknowledgment, and no guarantee of delivery. It is like sending a postal letter — you drop it in the post box and hope it reaches.

- NO connection setup needed
- Data sent directly
- No acknowledgment
- No guarantee of delivery
- Faster (no setup time)
- Less reliable

REAL EXAMPLES

(1) UDP protocol — used for video calls, gaming, live streaming · (2) Postal letter — drop in postbox, no guarantee of delivery · (3) SMS text message — send and hope it reaches · (4) DNS queries — quick lookup, no connection.

EXAM Q&A

Q: What is connectionless service? **A:** Service where data is sent without establishing connection first. Faster but unreliable. Example: UDP.

NO HANDSHAKE · UDP / POSTAL MAIL



Sender

Drops packets, hopes they arrive

NO CONNECTION SETUP — NO ACK

P1 →

P2 →

P3 →

Receiver

Hopes to receive



Send and hope — no handshake, no guarantee

06-6 CHAPTER 6 - LESSON 6

Connection-Oriented vs Connectionless Service

Feature	Connection-Oriented	Connectionless
Connection Setup	Required before data transfer	Not required
Phases	3 phases (setup, transfer, release)	1 phase (just send)
Reliability	High (guaranteed delivery)	Low (no guarantee)
Speed	Slower (setup overhead)	Faster (no setup)
Order of Packets	Maintained	May arrive out of order
Acknowledgment	Yes (confirms delivery)	No
Error Handling	Yes (resends lost data)	Minimal or none
Bandwidth Use	More (overhead)	Less
Protocol Example	TCP	UDP
Real Analogy	Telephone call	Postal letter
Use Cases	Web, email, file transfer	Video calls, gaming, DNS

EXAM Q&A

Q: Differentiate connection-oriented and connectionless service (5 points). **A:** See table — setup, phases, reliability, speed, protocol example.

06·7 CHAPTER 6 - LESSON 7

Analogy: Phone Call vs Postal Mail

Easy way to remember connection-oriented vs connectionless.



Connection-Oriented = Phone Call

CONNECT · TALK · HANG UP

1. You dial the number (**setup connection**)
2. Other person picks up — connection established
3. Both talk — data transfer happens
4. Either hangs up — connection released

Used by: TCP, telephone, ATM transaction. Reliable but takes time to setup. transaction.



Connectionless = Postal Mail

WRITE · DROP · HOPE

1. You write a letter (**prepare data**)
2. Drop it in postbox (**send — no setup**)
3. Postman delivers (hopefully)
4. No confirmation of delivery

Used by: UDP, SMS, DNS. Fast but unreliable.

MEMORY TRICK

Phone call = connect first, talk later (connection-oriented). Letter = drop and forget (connectionless).

REV UNIT 1 REVISION

Unit 1 Quick Revision — All Key Terms in One Page

Last-minute revision — memorize these definitions.

◆ Network

01

2+ devices connected to share data.

❄ Node

02

Any device on a network.

🖨 Server

03

Computer that provides services to clients.

📄 Protocol

04

Rules for data communication.

🏠 LAN

05

Network within building/campus (1 km).

🌐 WAN

06

Network across countries (Internet).

✓ TCP

07

Reliable, connection-oriented protocol.

⚡ UDP

08

Fast, connectionless protocol.

👤 P2P

09

All peers equal, no central server.

TIP

Write these 9 definitions in your notebook today. They cover **70% of Unit 1 exam questions!**

Q UNIT 1 · EXAM PREP

Common Exam Questions — Unit 1

Most likely questions based on 15% weightage and past papers.

1. Define computer network. List its 5 uses. 3 marks
2. Differentiate LAN, MAN, WAN. 5 marks
3. Explain 5 types of servers with example. 5 marks
4. Define protocol. Explain TCP vs UDP. 5 marks
5. Differentiate P2P and Client-Server architecture. 5 marks
6. Explain simplex, half-duplex, full-duplex with example. 3 marks
7. Differentiate Internet, Intranet, Extranet. 5 marks
8. What is network simulator? Why is it used? 2 marks
9. Differentiate point-to-point and multipoint configuration. 3 marks
10. Differentiate connection-oriented and connectionless service. 5 marks

STRATEGY These 10 questions cover **~41 marks**. Master them and you will score **12+ out of 15!**



Thank You!

That completes Unit 1 — Introduction to Computer Networks

1 Revise the 9 key terms slide.

2 Practice the 10 exam questions.

3 Move to Unit 2 — The Reference Models (OSI & TCP/IP).
TCP/IP).

ALL THE BEST!

Remember: Life is a race. Agar tez nahi bhaagoge, toh koi tumhe kuchal ke aage nikal jaayega.